

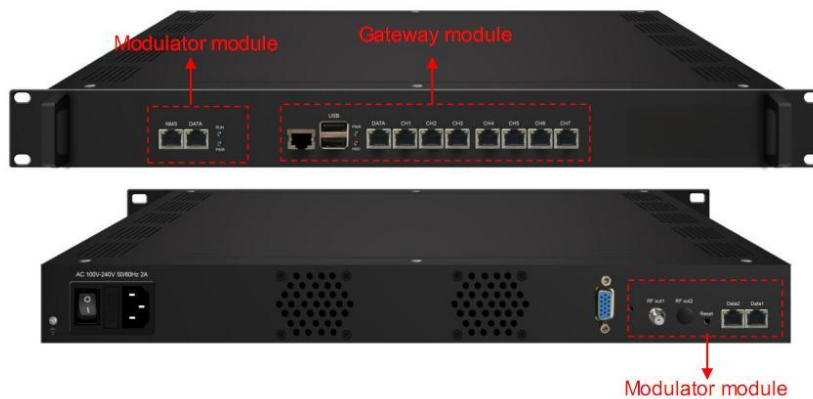
FMUSER FUQ-4619B IPTV Gateway with 8 IP Data Ports and Modulator 16 DVB-C-8 DVB-T-6 ISDBT RF Outputs IP to RF Conversion DVB-C-T-ISDBT-ATSC Standards Web NMS Management for CATV Headends and IPTV Systems



HTTP/UDP/RTP/RTSP/HLS



DVB-C/T/ISDBT/ATSC RF Out



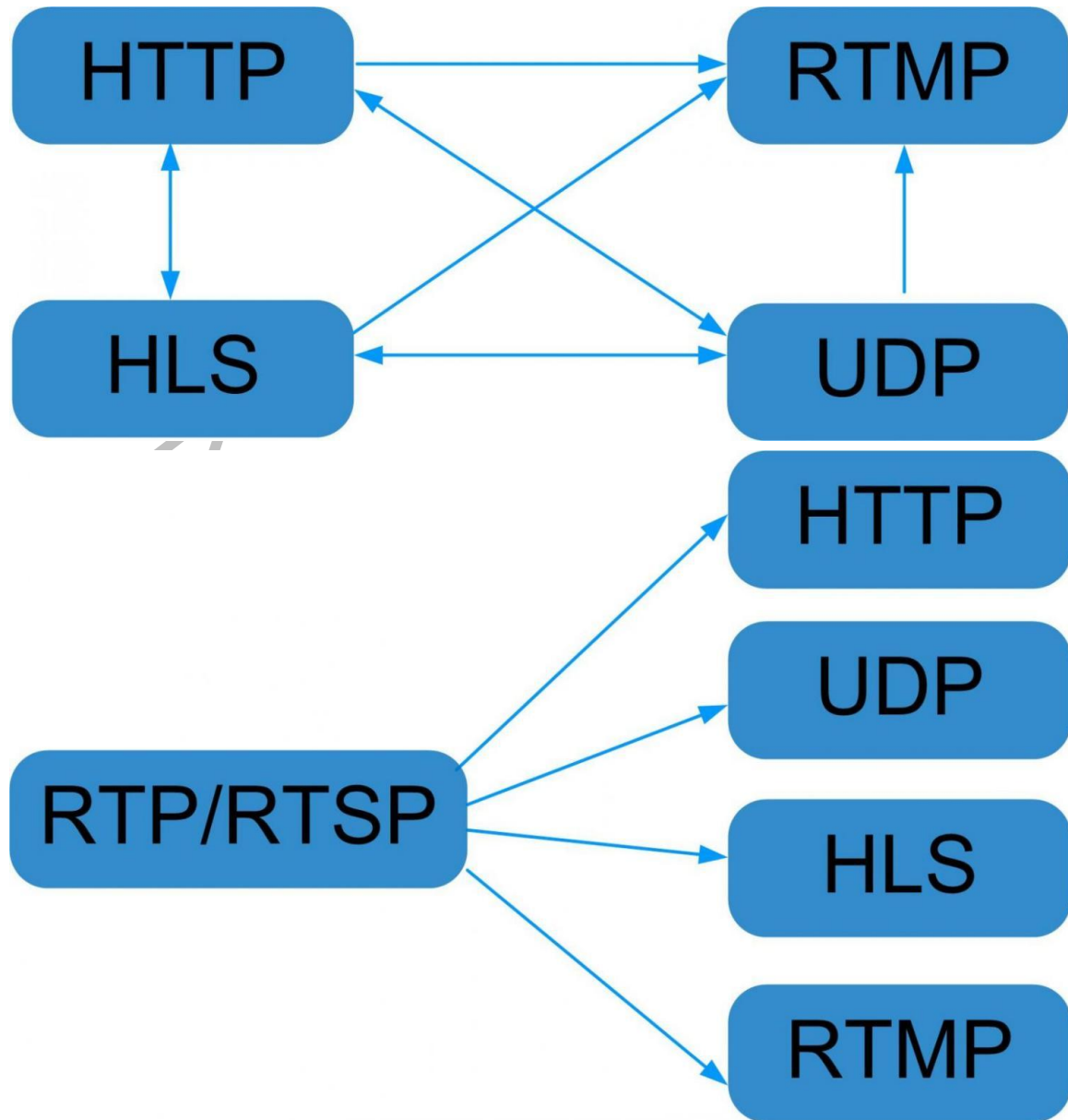
Product overview

The FMUSER FUQ-4619B combines an IPTV gateway with 8 data ports and a modulator for direct IP-to-RF conversion. It processes IP streams from protocols like HTTP/UDP/RTP to RF outputs compliant with DVB-C/T/ISDBT/ATSC standards. Managed via web-based NMS, this solution is ideal for CATV headends and IPTV/OTT systems for efficient program redistribution.

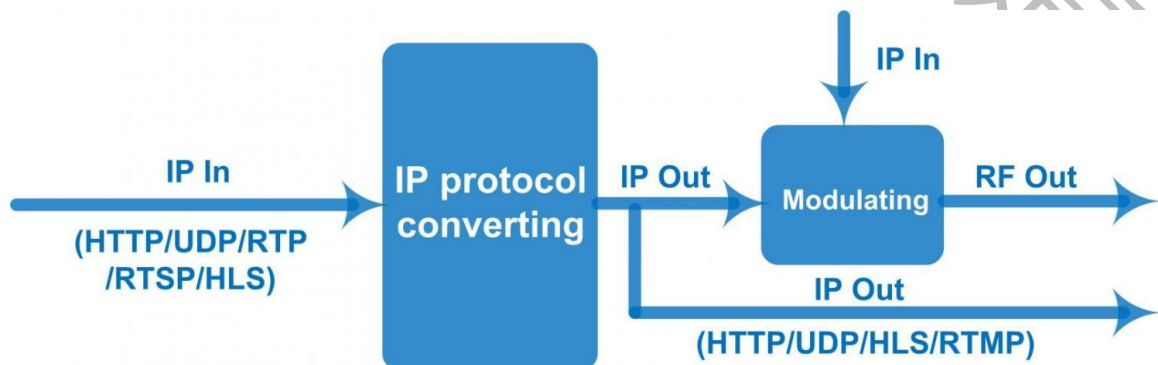
Key feature

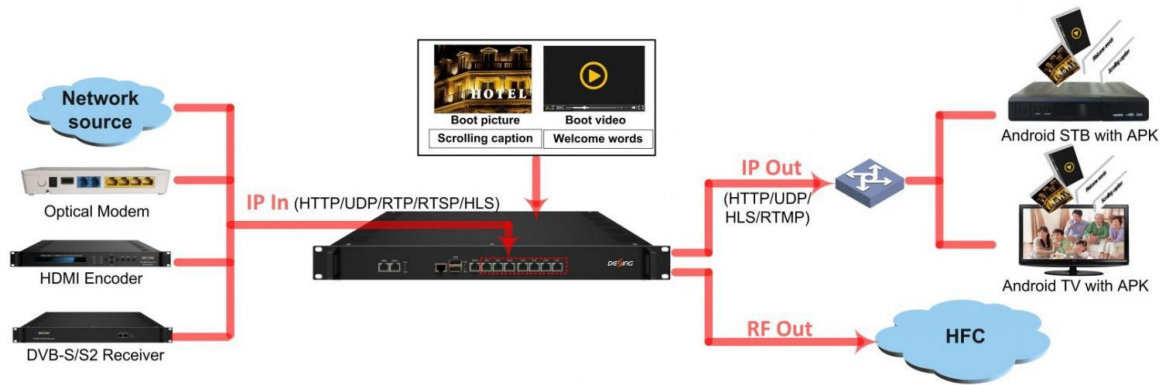
Enables centralized control via web-based NMS and TS file upload.

IP Protocol Conversion



Principle Chart





Scrolling caption/welcome words/boot picture/boot video is only applicable to IP out application and the STB/Android TV must be installed Dexin IPTV APK

Specifications

IPTV gateway module

Input	IP input thru CH 1-7(1000M) over HTTP, UDP(SPTS), RTP(SPTS), RTSP (over UDP, payload: mpeg TS) and HLS
Input	TS files uploading through Web management
IP output	IP out thru DATA port (1000M) over HTTP (Unicast), UDP(SPTS, Multicast) HLS and RTMP (Program source should be H.264 and AAC encoding) IP out thru CH 1-7(1000M) over HTTP/ HLS/RTMP (Unicast)
System	Memory: 4G
System	Solid-State Disk(SSD): 16G
System	Channel switching time with FMUSER' STB: HTTP (1-3s), HLS (0.4-0.7s)
System	Support adding scrolling caption, welcome words, boot picture and boot video (this function is only applicable to IP out application and the STB/Android TV must be installed FMUSER IPTV APK)
System	Support downloading FMUSER IPTV APK directly from this module
System	Play programs with APK downloaded android STB and TV, maximum 150 terminals

System	Support about 80 HD/SD programs (Bitrate: 2Mbps) When HTTP/RTP/RTSP/HLS is converted into UDP (Multicast), the actual application shall prevail, and suggest maximum 80% CPU utilization
System	web-based NMS management thru module's DATA port

Modulator module (Taking FMUSER's DX316/DX308T/DX306I/DX308AT modulator module as examples)

Input	DX316 (DVB-C)	512 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	512 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	512 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	512 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	512 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port
Input	DX308T (DVB-T)	256 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	256 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	256 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	256 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	256 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port
Input	DX306I (ISDBT)	192 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	192 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	192 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	192 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	192 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port
Input	DX308AT (ATSC)	256 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	256 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	256 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	256 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port	256 IP (MP TS/SPTS) input over UDP/RTP, 2 100/1000M Ethernet Port
Multiplexing		DVB-C	DVB-C	DVB-T/ ATSC	DVB-T/ ATSC	ISDBT

Multiplexing	Input Channel	512	512	256	256	192
Multiplexing	Output Channel	16	16	8	8	6
Multiplexing	Max PIDs	180 per channel	180 per channel	180 per channel	180 per channel	180 per channel
Multiplexing	Functions	PID remapping(auto/manually optional)	PID remapping(auto/manually optional)	PID remapping(auto/manually optional)	PID remapping(auto/manually optional)	PID remapping(auto/manually optional)
Multiplexing	Functions	PCR accurate adjusting	PCR accurate adjusting	PCR accurate adjusting	PCR accurate adjusting	PCR accurate adjusting
Multiplexing	Functions	PSI/SI table automatically generating	PSI/SI table automatically generating	PSI/SI table automatically generating	PSI/SI table automatically generating	PSI/SI table automatically generating
Scrambling Parameters (for DVB-C)	Max simulcrypt CA	4	4	4	4	4
Scrambling Parameters (for DVB-C)	Scramble Standard	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197
Scrambling Parameters (for DVB-C)	Connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection
Modulation Parameters	DVB-C		J.83A	J.83A	J.83B	J.83B
Modulation Parameters	DVB-C	Constellation	16/32/64/128/256QAM	16/32/64/128/256QAM	64QAM/256QAM	64QAM/256QAM
Modulation Parameters	DVB-C	Bandwidth	8M	8M	6M	6M
Modulation Parameters	DVB-C	Standard	EN300 429/ITU-T J.83A/B(DVB-C)	EN300 429/ITU-T J.83A/B(DVB-C)	EN300 429/ITU-T J.83A/B(DVB-C)	EN300 429/ITU-T J.83A/B(DVB-C)

Modulation Parameters	DVB-C	Symbol rate	5.0~7.0Mps, 1ksps stepping	5.0~7.0Mps, 1ksps stepping	5.0~7.0Mps, 1ksps stepping	5.0~7.0Mps, 1ksps stepping
Modulation Parameters	DVB-C	QAM channel	16 non-adjacent carrier outputs within 192M bandwidth	16 non-adjacent carrier outputs within 192M bandwidth	16 non-adjacent carrier outputs within 192M bandwidth	16 non-adjacent carrier outputs within 192M bandwidth
Modulation Parameters	DVB-C	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation Parameters	DVB-C	RF output level	-20~+10dBm, 0.1dB step	-20~+10dBm, 0.1dB step	-20~+10dBm, 0.1dB step	-20~+10dBm, 0.1dB step
Modulation Parameters	DVB-C	MER	≥ 40dB	≥ 40dB	≥ 40dB	≥ 40dB
Modulation Parameters	DVB-C	FEC	RS (204, 188)	RS (204, 188)	RS (204, 188)	RS (204, 188)
Modulation Parameters	DVB-T	Standard	ETSI EN300 744	ETSI EN300 744	ETSI EN300 744	ETSI EN300 744
Modulation Parameters	DVB-T	Constellation	QPSK/16QAM/64QAM	QPSK/16QAM/64QAM	QPSK/16QAM/64QAM	QPSK/16QAM/64QAM
Modulation Parameters	DVB-T	Bandwidth	6/7/8 MHz	6/7/8 MHz	6/7/8 MHz	6/7/8 MHz
Modulation Parameters	DVB-T	Trans mode	2K/4K/8K	2K/4K/8K	2K/4K/8K	2K/4K/8K
Modulation Parameters	DVB-T	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation Parameters	DVB-T	MER	≥ 40dB	≥ 40dB	≥ 40dB	≥ 40dB
Modulation Parameters	DVB-T	RF frequency	50~960MHz, 1kHz stepping	50~960MHz, 1kHz stepping	50~960MHz, 1kHz stepping	50~960MHz, 1kHz stepping
Modulation Parameters	DVB-T	RF output Level	-20~+10dBm, 0.5dB stepping	-20~+10dBm, 0.5dB stepping	-20~+10dBm, 0.5dB stepping	-20~+10dBm, 0.5dB stepping

Modulation Parameters	DVB-T	RF out channel	8 non-adjacent carrier outputs within 192M bandwidth	8 non-adjacent carrier outputs within 192M bandwidth	8 non-adjacent carrier outputs within 192M bandwidth	8 non-adjacent carrier outputs within 192M bandwidth
Modulation Parameters	ISDBT	Standard	ARIB STD-B31	ARIB STD-B31	ARIB STD-B31	ARIB STD-B31
Modulation Parameters	ISDBT	Bandwidth	6M	6M	6M	6M
Modulation Parameters	ISDBT	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation Parameters	ISDBT	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4
Modulation Parameters	ISDBT	Transmission Mode	2K, 4K, 8K	2K, 4K, 8K	2K, 4K, 8K	2K, 4K, 8K
Modulation Parameters	ISDBT	Code rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation Parameters	ISDBT	MER	≥ 40dB	≥ 40dB	≥ 40dB	≥ 40dB
Modulation Parameters	ISDBT	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation Parameters	ISDBT	RF output level	-20dBm~+10dBm, 0.1dB stepping	-20dBm~+10dBm, 0.1dB stepping	-20dBm~+10dBm, 0.1dB stepping	-20dBm~+10dBm, 0.1dB stepping
Modulation Parameters	ISDBT	RF out	6 non-adjacent carrier outputs within 192M bandwidth	6 non-adjacent carrier outputs within 192M bandwidth	6 non-adjacent carrier outputs within 192M bandwidth	6 non-adjacent carrier outputs within 192M bandwidth
Modulation Parameters	ATSC	Standard	ATSC A/53	ATSC A/53	ATSC A/53	ATSC A/53
Modulation Parameters	ATSC	Bandwidth	6M	6M	6M	6M

Modulation Parameters	ATSC	Constellation	8VSB	8VSB	8VSB	8VSB
Modulation Parameters	ATSC	FEC	RS(208 18 8)+Trellis	RS(208 18 8)+Trellis	RS(208 18 8)+Trellis	RS(208 18 8)+Trellis
Modulation Parameters	ATSC	MER	≥40dB	≥40dB	≥40dB	≥40dB
Modulation Parameters	ATSC	RF frequency	50~960MHz, 1kHz step	50~960MHz, 1kHz step	50~960MHz, 1kHz step	50~960MHz, 1kHz step
Modulation Parameters	ATSC	RF output level	-20dBm~+10dBm, 0.5dB stepping	-20dBm~+10dBm, 0.5dB stepping	-20dBm~+10dBm, 0.5dB stepping	-20dBm~+10dBm, 0.5dB stepping
Modulation Parameters	ATSC	RF out	8 non-adjacent carrier outputs within 192M bandwidth	8 non-adjacent carrier outputs within 192M bandwidth	8 non-adjacent carrier outputs within 192M bandwidth	8 non-adjacent carrier outputs within 192M bandwidth
TS output	16 (DVB-C)/8(DVB-T/ATSC)/6(ISDBT) IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M Ethernet Ports (Data 1/2)	16 (DVB-C)/8(DVB-T/ATSC)/6(ISDBT) IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M Ethernet Ports (Data 1/2)	16 (DVB-C)/8(DVB-T/ATSC)/6(ISDBT) IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M Ethernet Ports (Data 1/2)	16 (DVB-C)/8(DVB-T/ATSC)/6(ISDBT) IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M Ethernet Ports (Data 1/2)	16 (DVB-C)/8(DVB-T/ATSC)/6(ISDBT) IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M Ethernet Ports (Data 1/2)	16 (DVB-C)/8(DVB-T/ATSC)/6(ISDBT) IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M Ethernet Ports (Data 1/2)
System	web-based NMS management thru module's NMS port	web-based NMS management thru module's NMS port	web-based NMS management thru module's NMS port	web-based NMS management thru module's NMS port	web-based NMS management thru module's NMS port	web-based NMS management thru module's NMS port
General		Demission		482mm×324mm×44mm (W×L×H)		

General	Temperature	0~45■(operation), -20~80■(storage)
General	Power Supply	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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